Aman Monga

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Education

University of Victoria — Bachelor of Software Engineering (2020–2026)

- *Degree completed via co-operative education program, combining full-time study with industry placements; employment gaps reflect academic terms*

AWS - Certified Cloud Practitioner (in progress)

Work Experience

Developer (*Contractor***)** — Livmora Solutions, Victoria, BC *May 2026 – Present*

- Sole developer at a software contracting startup, identifying and pitching technology solutions to local businesses (pubs, dental offices, trades companies) — from website builds, AI solutions and automated workflows that save them hours of manual admin
- Led a full digital presence overhaul for The Prairie Inn Pub - a historic neighbourhood pub established in 1859 and Vancouver Island's first officially designated neighbourhood pub (1974) — covering website design, logo redesign, and brand identity, now live in production handling ~2,000 requests/day
- Restructuring software architecture for Kintec Footwear and Orthotics, an 85-employee, 7-locations BC footwear + orthotics retailer (~$24.6M/¥2.7 billion annual revenue), including data lake implementation and workflow automation

Product Developer (*Full time intern, 5-month term***)** — SSI, Victoria, BC *Sep 2025 – Jan 2026*

- Contributed to the development of ShipConstructor, the industry-leading AutoCAD-based 3D CAD/CAM software for ship and offshore structure design, at SSI — a 35+ year-old software company (~150 employees, ~$15-27M/¥1.65–2.97 billion annual revenue) with offices in Canada and the United States, serving shipbuilders globally
- Worked full-time as an embedded member of the engineering team, carrying the same day-to-day responsibilities as permanent engineers
- Developed backend modules and database access layers using C#, .NET, C++, and Entity Framework Core, along with system integration services
- Built interoperability layers for efficient platform communication and data exchange
- Refactored legacy WinForms/MFC interfaces into modern WPF following MVVM architecture
- Collaborated in agile delivery cycles including planning, reviews, and production support

Full Stack Developer (*Full time contract, 14-month term***)** — Quartech**,** Victoria, BC *May 2023 - June 2024*

- Developed features for Property Information Management System (PIMS) at Quartech — a 40+ year-old IT consulting firm (~150-250 employees, $25M-50M/~¥2.5-5.5B annual revenue) with offices in Canada and the United States— building a platform used by the BC Ministry of Transportation to manage provincial real property assets
- Worked full-time as an embedded member of the engineering team, carrying the same day-to-day responsibilities as permanent developers, including direct client interaction
- Collaborated with clients to gather requirements and present project deliverables
- Built frontend components using React, Vue.js, and TypeScript, with backend support in C#/.NET
- Wrote SQL queries for data processing and resolved Docker and Keycloak-related issues

Backend Python Developer (*Full time intern, 8-month term***)** — CFIA *May 2022 - Dec 2022*

- Developed RESTful APIs using aioHTTP for Virtool, an open-source platform for analyzing next-generation sequencing (NGS) data to detect viral infections — used in a specialized,

high-containment research environment (Canadian Food Inspection Agency (CFIA) operates Canada's only Containment Level 4 animal health lab)
- Designed and implemented a WebSocket service for real-time messaging using Redis Pub/Sub
- Built database functionality for PostgreSQL and MongoDB using SQLAlchemy and PyMongo
- Implemented unit tests with Pytest to improve reliability and maintainability

Recent Projects

AI Engineer (*Short term contract***)** — Sidian.io *April 2026 - May 2026*

- Built a privacy law Q&A system independently for Sidian Technologies, an early-stage startup building AI-powered document compliance tooling for law firms
- Designed a two-stage RAG pipeline with hybrid dense/sparse search and neural reranking to answer compliance questions
- Implemented the hybrid search with BGE-M3 embeddings and Qdrant, followed by cross-encoder reranking (bge-reranker-large) and grounded generation via a locally-hosted LLM (llama3.1:8b via Ollama), served through a FastAPI backend and React/Vite frontend

System Design & DevOps — Independent Technical Development *In progress*

- Currently Containerizing the above mentioned RAG application using Docker Compose across four services, to better understand docker tooling
- Setting up a GitHub Actions CI/CD pipeline for automated Docker image builds and deployment
- Studying system design fundamentals like load balancing, Redis caching, and cloud infrastructure

Technical Skills

- **Languages:** C#, Python, Java, C/C++
- **Frontend:** React, Vue.js, HTML, CSS, JavaScript, TypeScript
- **Backend:** .NET, FastAPI, aioHTTP
- **AI/Data:** TensorFlow, scikit-learn, Pandas, LangChain
- **Databases:** PostgreSQL, ChromaDB, SQL, Qdrant
- **Tools:** Docker, Git, GitHub Actions, Azure DevOps, AWS
- **Unit Testing:** Pytest, Jest, xUnit
- **Other:** Microsoft Dynamics 365 (X++, SSRS), Linux/Unix